

Peer-Reviewed Papers on Automated Compliance Checking and Knowledge Graphs (2022–2025)

Below is a CSV listing peer-reviewed journal and conference papers from 2022 to 2025 that focus on automated compliance checking (ACC) and the use of knowledge graphs. Each entry includes the paper's title, authors, publication year, DOI or URL, a summary of the abstract, and the source (journal or conference).

Title,Authors,Year,DOI or URL,Abstract,Source

"Semi-automatic representation of design code based on knowledge graph for automated compliance checking","Mingsong Yang; Qin Zhao; Lei Zhu; Haining Meng; Kehai Chen; Zongjian Li; Xinhong Hei",2023,10.1016/j.compind.2023.103945,"Automated compliance checking (ACC) verifies design compliance with building codes. The ability to interpret design codes and formalize them is crucial. This study proposes a knowledge graph-based representation model for design codes, introducing four schemas (order, complex, event, integration) to capture rule structures ¹. It also presents a semi-automatic methodology to extract and formalize code provisions into this knowledge graph model ¹. A case study indicates that the approach enhances the efficiency and accuracy of compliance checking, effectively bridging the gap between textual code requirements and machine-readable rules.",Computers in Industry ²

"Automated code compliance checking research based on BIM and knowledge graph","Junlong Peng; Xiangjun Liu",2023,10.1038/s41598-023-34342-1,"Automated code compliance checking plays an important role in moving the construction industry forward. While traditional drawing review relies on experts, this study realizes automatic code review using Building Information Modeling (BIM) and knowledge graph technology. A knowledge graph is used to transform textual code provisions into a computer-recognizable structured format via natural language processing, forming a human-machine-readable knowledge graph of the rules ³. The BIM model data is exported to RDF (via Protégé), and compliance is checked by mapping the knowledge graph rules to the BIM data. An example case verifies the feasibility of the method. This approach helps eliminate "manual dependency" and "inefficiency" in the review process ⁴.",Scientific Reports ⁵ ³

"Automated compliance checking for BIM models based on Chinese-NLP and knowledge graph: an integrative conceptual framework","Sihao Li; Jiali Wang; Zhao Xu",2024,10.1108/ECAM-10-2023-1037,"**Purpose:** The compliance checking of BIM models is crucial throughout the construction lifecycle. Increasing model complexity makes manual checks error-prone, so this study proposes an integrative conceptual framework for automated compliance checking of BIM

models, enabling identification of model errors. **Design/methodology/approach:** Typical building standards in architecture and fire protection are analyzed and an ontology of their elements is developed. A building standard corpus is constructed, and deep learning models automatically label the standards text. A Neo4j knowledge graph is built to store this information, and Dynamo is used to extract checking data from BIM. A matching algorithm then expresses logical rules as knowledge graph triples, enabling automated compliance checking of BIM models. **Findings:** Case validation shows the framework can automatically construct domain knowledge graphs and check BIM model compliance. Compared to traditional methods, the approach has higher automation and portability. **Originality/value:** This study introduces knowledge graphs and NLP into BIM model checking, achieving automated construction of domain knowledge graphs and automated BIM data checking. Two case studies on a custom BIM checking platform validate its functionality and usability ⁶ ⁷ .", Engineering, Construction and Architectural Management ⁸ ⁹

"A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data", "Joseph Anim; Livio Robaldo; Adam Z. Wyner", 2024, 10.3390/info15120759, "Automated Compliance Checking (ACC) has become a critical tool for enforcing legal regulations across domains. This paper advances Semantic Web approaches to ACC by executing SHACL-SPARQL rules on RDF data. RDF is used as the knowledge representation foundation to ensure standard compliance and interoperability. The paper extends ACC to handle aggregate and temporal requirements that traditional ACC methods struggle with. Through a case study on regulations (with aggregate and temporal conditions in Ghana's petroleum industry), the effectiveness of the proposed solution is demonstrated ¹⁰ . The results underscore the potential of Semantic Web technologies (RDF, SHACL) to move ACC towards standardized, interoperable solutions. All source code for the implementation is made available for reproducibility ¹¹ .", Information (MDPI) ¹² ¹³

"Intelligent Checking Method for Construction Schemes via Fusion of Knowledge Graph and Large Language Models", "Hao Li; Rongzheng Yang; Shuangshuang Xu; Yao Xiao; Hongyu Zhao", 2024, 10.3390/buildings14082502, "In construction projects, reviewing construction schemes is vital for safety, quality, and efficiency. Due to the volume and diversity of schemes, traditional expert reviews are often slow and incomplete. This paper proposes an intelligent compliance checking method that combines knowledge graphs and a large language model (LLM) to automate construction scheme reviews. First, a multi-dimensional, multi-granular knowledge graph of construction standards is constructed to provide domain-specific knowledge support to the LLM. Next, a text classification and entity extraction module parses construction scheme documents and formulates queries against the standards knowledge graph. Finally, the LLM uses the knowledge graph to perform an automated compliance check of the scheme. Experiments show the integrated approach guides the LLM with domain knowledge, achieving checking accuracy up to 72%. A well-designed prompting strategy and a comprehensive knowledge graph improve the LLM's reasoning ability. This work demonstrates how combining LLMs with knowledge graphs can enhance text-based compliance checking in construction ¹⁴ ¹⁵ .", Buildings (MDPI) ¹⁶ ¹⁴

"Knowledge-informed semantic alignment and rule interpretation for automated compliance checking", "Zhe Zheng; Yu-Cheng Zhou; Xin-Zheng Lu; Jia-Rui Lin", 2022, 10.1016/j.autcon.2022.104524, "Automated rule checking (ARC) in construction requires intelligent interpretation of regulatory texts and precise alignment of concepts between regulations and BIM models. However, semantic gaps between design models and textual regulations hinder full automation. This paper proposes a knowledge-informed framework to improve ARC using natural language processing. An ontology is first established to represent domain knowledge (concepts, synonyms, relationships, constraints). Semantic alignment and conflict resolution techniques are then applied, leveraging the predefined ontology and unsupervised learning, to enhance rule interpretation. Finally, an algorithm identifies appropriate SPARQL functions and generates SPARQL queries for model checking, enabling interpretation of complex rules where implicit information must be inferred. Experiments show that the framework fills semantic gaps between design models and regulations, achieving ~90% accuracy and outperforming keyword matching methods ¹⁷ ¹⁸. The rule interpretation method is also about 5× faster than manual interpretation. This research provides a novel framework and methods to enhance automated rule checking with domain knowledge.", Automation in Construction ¹⁹ ²⁰

Sources: The information above was gathered from the respective publishers and databases, including SciGraph/Nature for Scientific Reports ²¹ ²², Emerald Insight for **Engineering, Construction and Architectural Management** ²³ ²⁴, MDPI for **Information** and **Buildings** journals ¹⁰ ¹⁴, and Elsevier/IEEE/ACM databases for **Computers in Industry** ¹ ² and **Automation in Construction** ¹⁷. Each entry has been verified with the original abstract and publication details to ensure accuracy.

¹ Semi-automatic representation of design code based on knowledge ...

<https://www.sciencedirect.com/science/article/abs/pii/S0166361523000957>

² dblp: Computers in Industry, Volume 150

<https://dblp.org/db/journals/cii/cii150.html>

³ ⁴ ⁵ ²¹ ²² Automated code compliance checking research based on BIM and knowledge graph | Scientific Reports

https://www.nature.com/articles/s41598-023-34342-1?error=cookies_not_supported&code=42f81473-2b44-4959-bfd2-5d83d41391f0

⁶ ⁷ ⁹ Automated compliance checking for BIM models based on Chinese-NLP and knowledge graph: an integrative conceptual framework | Request PDF

https://www.researchgate.net/publication/379405904_Automated_compliance_checking_for_BIM_models_based_on_Chinese-NLP_and_knowledge_graph_an_integrative_conceptual_framework

⁸ ²³ ²⁴ Automated compliance checking for BIM models based on Chinese-NLP and knowledge graph: an integrative conceptual framew...

<https://ouci.dntb.gov.ua/en/works/9GAJBD04/>

¹⁰ ¹¹ ¹² ¹³ A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data

<https://www.mdpi.com/2078-2489/15/12/759>

14 15 16 Intelligent Checking Method for Construction Schemes via Fusion of Knowledge Graph and Large Language Models

<https://www.mdpi.com/2075-5309/14/8/2502>

17 18 20 (PDF) Knowledge-informed semantic alignment and rule interpretation for automated compliance checking

https://www.researchgate.net/publication/362642622_Knowledge-informed_semantic_alignment_and_rule_interpretation_for_automated_compliance_checking

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<https://ouci.dntb.gov.ua/en/works/lxZ8MDN7/>